RAMIRO YAIR ROMÁN TREJO

Software Engineer | Junior Backend & DevOps Engineer

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PROFESSIONAL SUMMARY

Software Engineer specialized in **Backend Architecture (Python, Django, FastAPI)**, **DevOps Infrastructure (Docker Compose, Jenkins, Linux, AWS EC2, Nginx)**, and **Observability (Prometheus, Grafana)**, backed by rigorous **Computer Science foundations (low-level Java data structures, graph theory, compilers, and cryptography)** and **Computer Vision / AI pipelines (YOLOv8, PyTorch)**. Proven track record delivering production platforms in industrial manufacturing (**Compañía Cervecera de Zacatecas / Grupo Modelo – AB InBev**), leading institutional systems (**UAZ**), and enforcing software quality assurance (**IV&V, BDD testing, and Humphrey's PSP**).

TECHNICAL SKILLS

Languages & CS Foundations: Python 3.x, Java (Binary Search Trees, Heaps, Directed Graphs, Dijkstra's Algorithm, Stack Compilers/Parsers), Kotlin, Dart, SQL, C#, PHP, C/C++.

Backend & Databases: Django 5.x, Django REST Framework (DRF), FastAPI (Async/Microservices), Spring Boot, Laravel 8, PostgreSQL 16, MariaDB/MySQL 8, MongoDB 6, SQLite/Room, Oracle PL/SQL.

DevOps, Cloud & Observability (SRE): Docker, Docker Compose (isolated meshes & healthchecks), Jenkins (`Jenkinsfile`), GitHub Actions, AWS EC2, Linux (Debian/Ubuntu, Bash), Nginx Proxy Manager (SSL), Prometheus, Grafana, Alertmanager, Node Exporter, cAdvisor, Blackbox Exporter.

AI, Cryptography & Mobile Development: Ultralytics YOLOv8/v5, PyTorch, AgML, Hidden Markov Models; Asymmetric RSA (TCP Sockets), RC4 Stream Cipher, Shannon Entropy; Flutter & Dart (SQLite FFI), Native Android (Kotlin, MVVM, Google Maps API).

Quality Assurance (QA), Testing & Methodologies: BDD Testing with Behave (Gherkin), Selenium WebDriver (Headless Firefox in CI), pytest, Flake8, Independent Verification & Validation (IV&V), Humphrey's PSP (SEI), Root Cause Analysis (RCA), Agile / Scrum.

WORK EXPERIENCE & TECHNICAL LEADERSHIP

Web Developer / Software Consultant — Compañía Cervecera de Zacatecas (Grupo Modelo / AB InBev) **Aug 2026 – Oct 2026**

- Architected and deployed a centralized web platform in **Django 5.0 and PostgreSQL 16** (containerized with **Docker Compose**) for auditing, tracking, and compliance of tooling and Good Operating Practices (GOPs) across the Packaging Department.
- Digitized production line inspection workflows and automated action-plan generation, eliminating paper logging lag and integrating industrial operational metrics (**MTTR and LEF**) with SharePoint and Power BI.

Technical Lead & Backend Developer — Universidad Autónoma de Zacatecas (UAZ) – Lic. en Lenguas Extranjeras **Nov 2025 – May 2026**

- Led end-to-end backend development of the academic curriculum tracking platform using **Python, Django, and MariaDB** over Debian Linux servers with Apache `mod_wsgi`.
- Engineered a declarative CI/CD pipeline in **Jenkins (`Jenkinsfile`)** featuring static linting (**Flake8**), **BDD acceptance testing (Behave + Headless Selenium)**, private Docker registry pushes, and automated **SSH deployment to AWS EC2** with automatic version rollback.

Quality Assurance Lead (QA Lead) & Android Developer — MovilSecure (4Nodos Collaborative Project) **Oct 2025 – Nov 2025**

- Directed **Independent Verification & Validation (IV&V)** procedures, specification audits, formal Design & Code Peer Reviews, and contributed to native **Kotlin development (MVVM architecture, Room SQLite, and Google Maps API)**.

Quality Assurance (QA) Inspector — Nagakura Engineering Works **Apr 2021 – Jul 2022**

- Supervised dimensional and material inspection protocols under strict manufacturing tolerances, assembly line defect containment, and **Root Cause Analysis (RCA)** for corrective actions and specification audits.

FEATURED ENGINEERING PROJECTS

- **Doctor Chile – AI Crop Pathology Detection (YOLO) & Mobile App (`doctor_chile_app` / `yolo_chili_training`):** Supervised training of **Ultralytics YOLOv8/v5 models in PyTorch** classifying 5 foliar diseases with AgML curation and visual attention heatmaps; cross-platform mobile app in **Flutter & Dart** with offline-first **SQLite FFI** persistence and **GitHub Actions** CI/CD.
- **Core Data Structures Engine, Dijkstra's Algorithm & Syntax Compiler in Java (`data-structures-and-algorithms-java`):** Built from scratch at memory level (zero `java.util` collections): Binary Search Trees (BST), Heaps, Expression Trees (Infix/Postfix), Directed Graphs with shortest-path routing (**Dijkstra**), stack-based syntax compiler, binary WAV audio signal processing, and 2D/3D image matrix algebra.
- **Polyglot Microservices Mesh (`examen1-deploy`) & Cloud Observability Stack (`Observabilidad`): Docker Compose** cluster orchestrating 4 stacks (**FastAPI + PostgreSQL**, **Spring Boot + MongoDB**, **Laravel 8 + MySQL**, and **React Admin**) with isolated networks and active healthchecks; cloud monitoring on **AWS EC2** with **Prometheus, Grafana, Alertmanager, Node Exporter, cAdvisor, and Blackbox Exporter**.
- **University Enrollment Platform (`UPN-SASCE`) & RSA Cryptosystem (`Criptografia`):** Academic enrollment system for **Universidad Pedagógica Nacional (UPN)** in **Django, PostgreSQL, Docker, and Jenkins** verified via **Behave BDD** tests; client-server asymmetric **RSA encryption over TCP Sockets** and **RC4 stream cipher** in Python.

EDUCATION & LANGUAGES

B.S. in Software Engineering — Universidad Autónoma de Zacatecas (UAZ) **2022 – Present (In Progress)**

Languages: Spanish (Native) • English (B1 / Technical Intermediate — Proficient in reading architecture RFCs, technical documentation, and engineering collaboration).